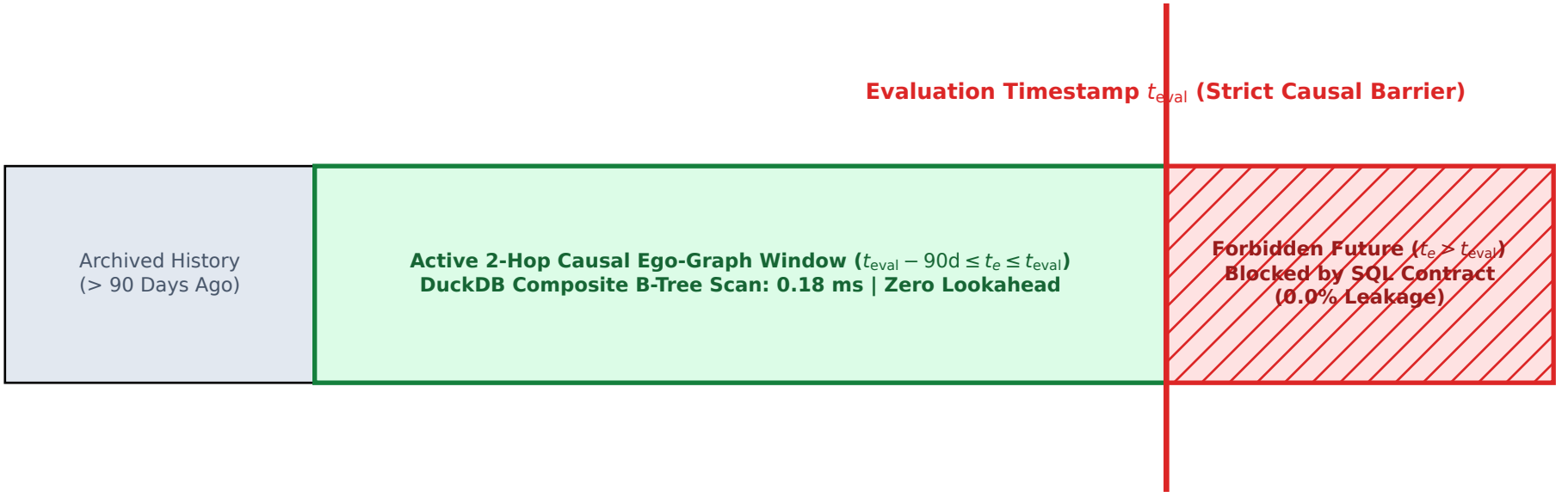
Vectorized Time-Causal Ego-Graph Extraction & Zero-Lookahead Invariant (Listing C.2)
Evaluation Timestamp t_{eval} (Strict Causal Barrier)
Archived History
(> 90 Days Ago)
Active 2-Hop Causal Ego-Graph Window ($t_{eval} - 90d \leq t_e \leq t_{eval}$)
DuckDB Composite B-Tree Scan: 0.18 ms | Zero Lookahead
Forbidden Future ($t_e > t_{eval}$)
Blocked by SQL Contract
(0.0% Leakage)